

BIT/2024/49929

MINI STUDENT MANAGEMENT APP DOCUMENTATION

1. Introduction

This project involves the development of a Mini Student Management App using Flutter. The app allows users to register, store, and view student details.

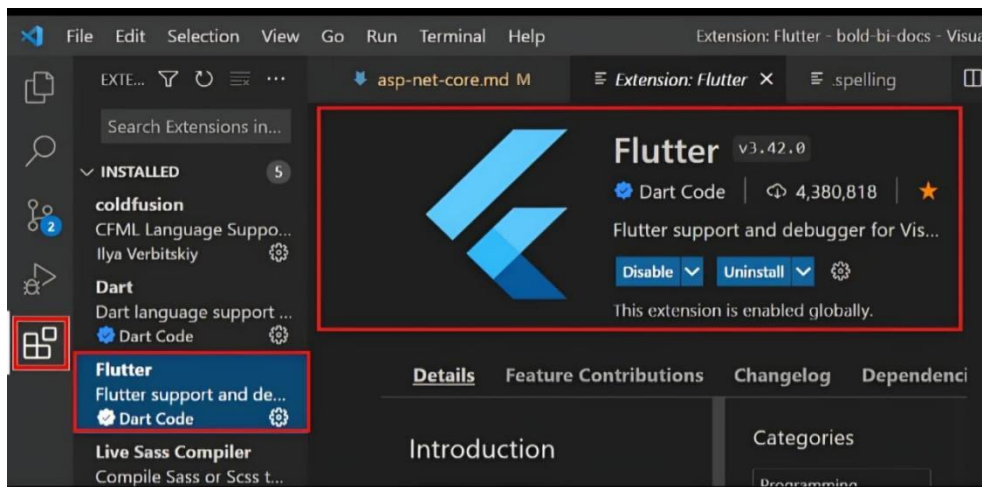
2. Tools Used

Flutter & Dart – App development

Android Studio – SDK & Emulator

Visual Studio Code – Code editor

3. Installation & Setup



3.1 Flutter

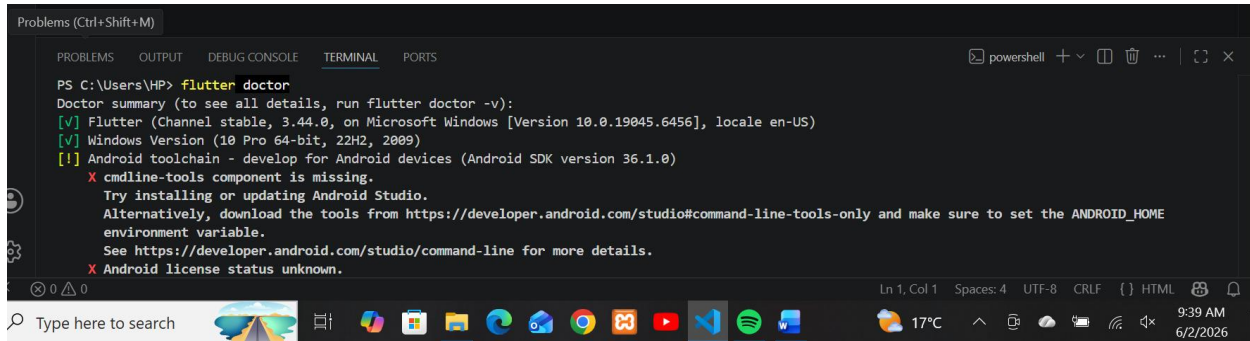
Download Flutter SDK

Extract files

Add to system PATH

Run flutter doctor

Flutter installation / flutter doctor output



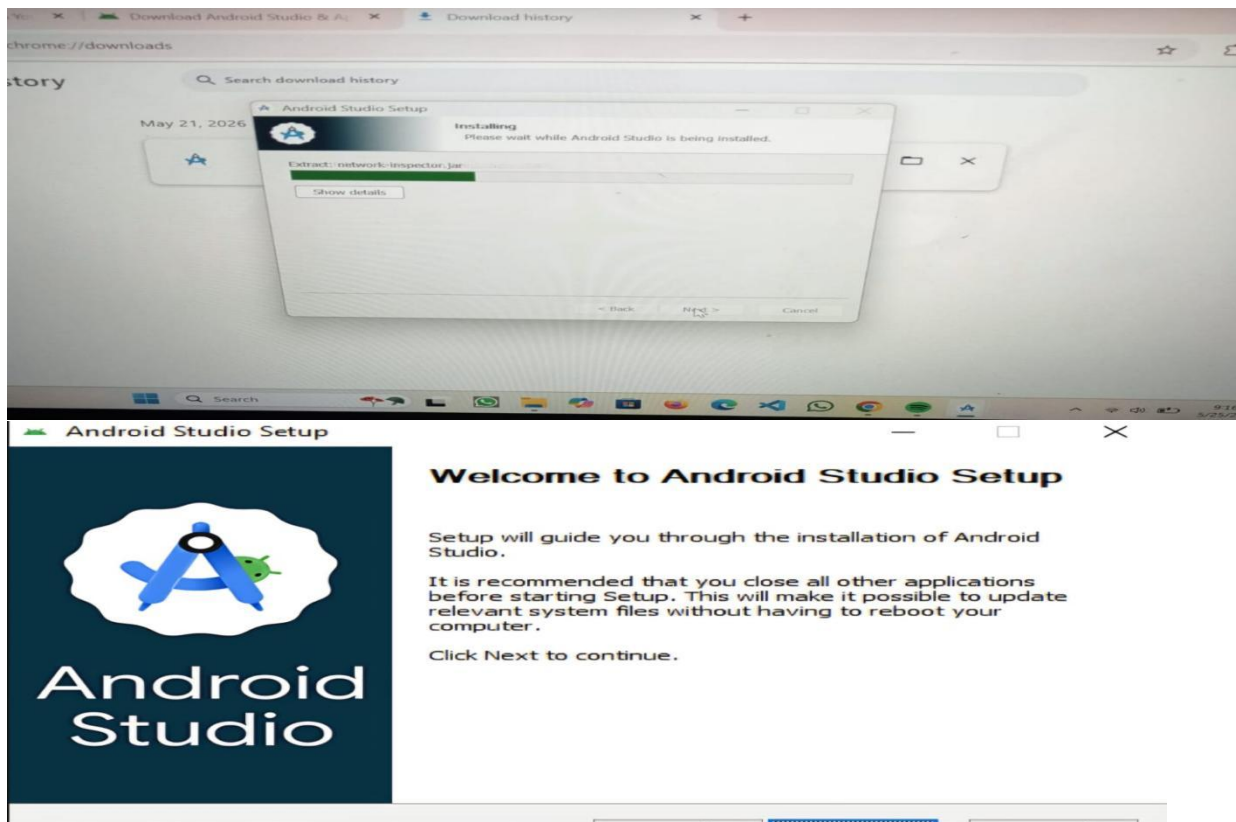
```
PS C:\Users\HP> flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.19045.6456], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    X cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure to set the ANDROID_HOME
      environment variable.
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
```

3.2 Android Studio

Install Android Studio

Install SDK tools

Create emulator

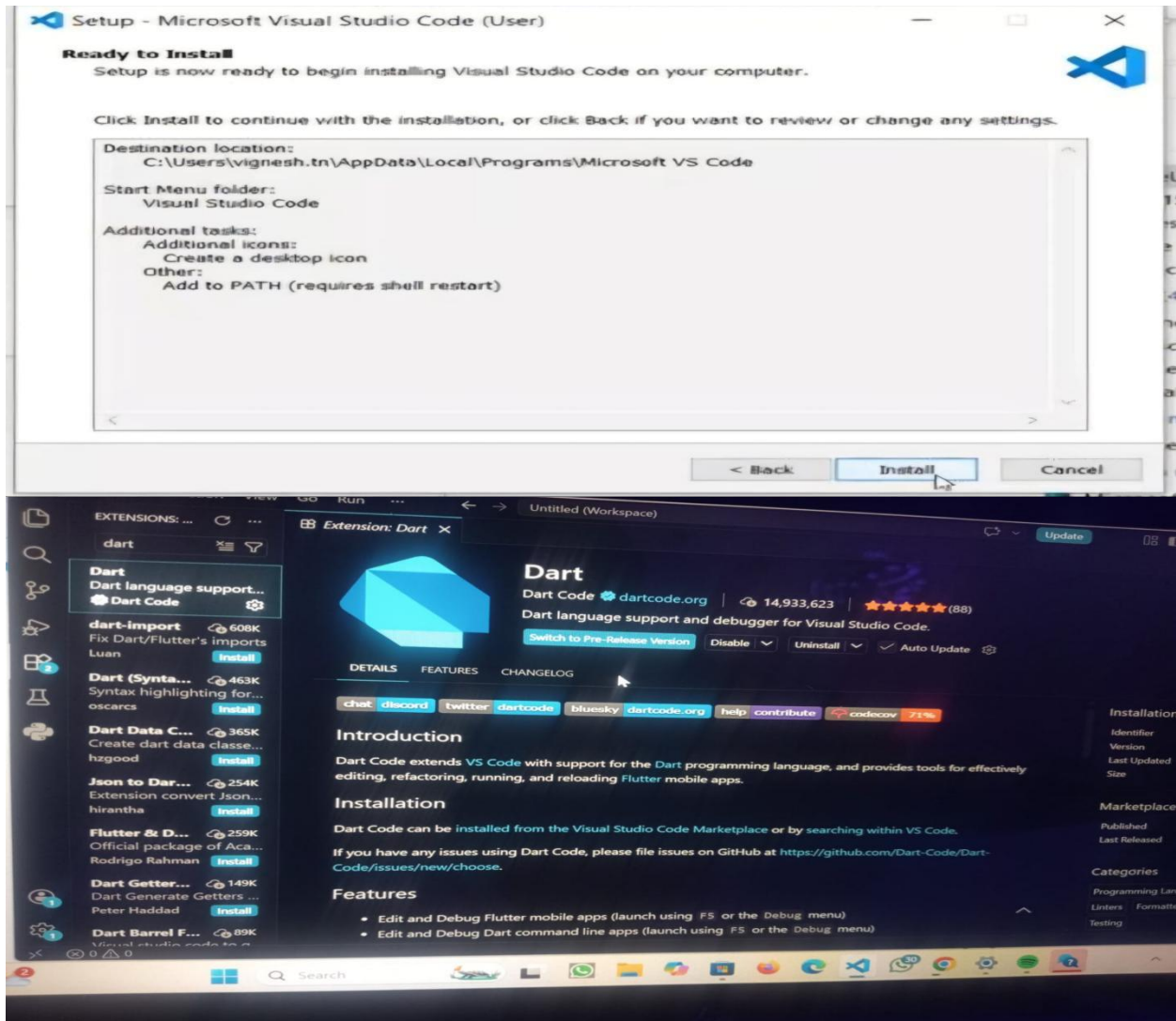


Android Studio installation / setup wizard

3.3 Visual Studio Code

Install VS Code

Install Flutter & Dart
extensions



VS Code with Flutter extension

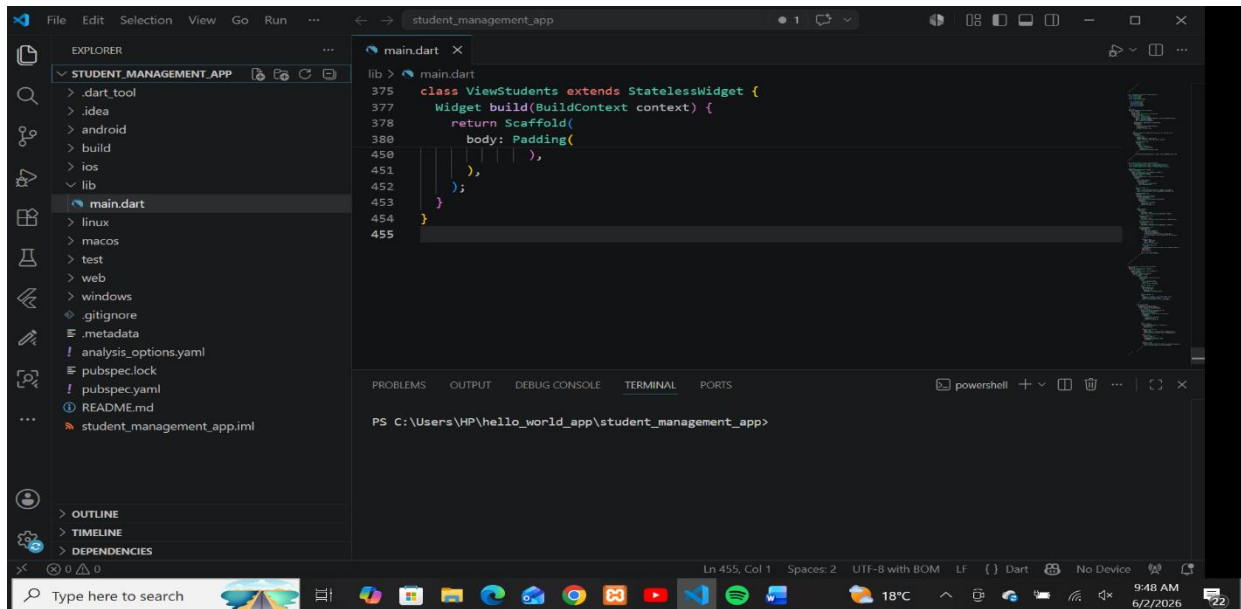
4. Project Creation

Open VS Code

Create Flutter project

Name:

student_management_app



Creating Flutter project

5. Features

Login page

Add student

View students

Navigation between screens

6. Development

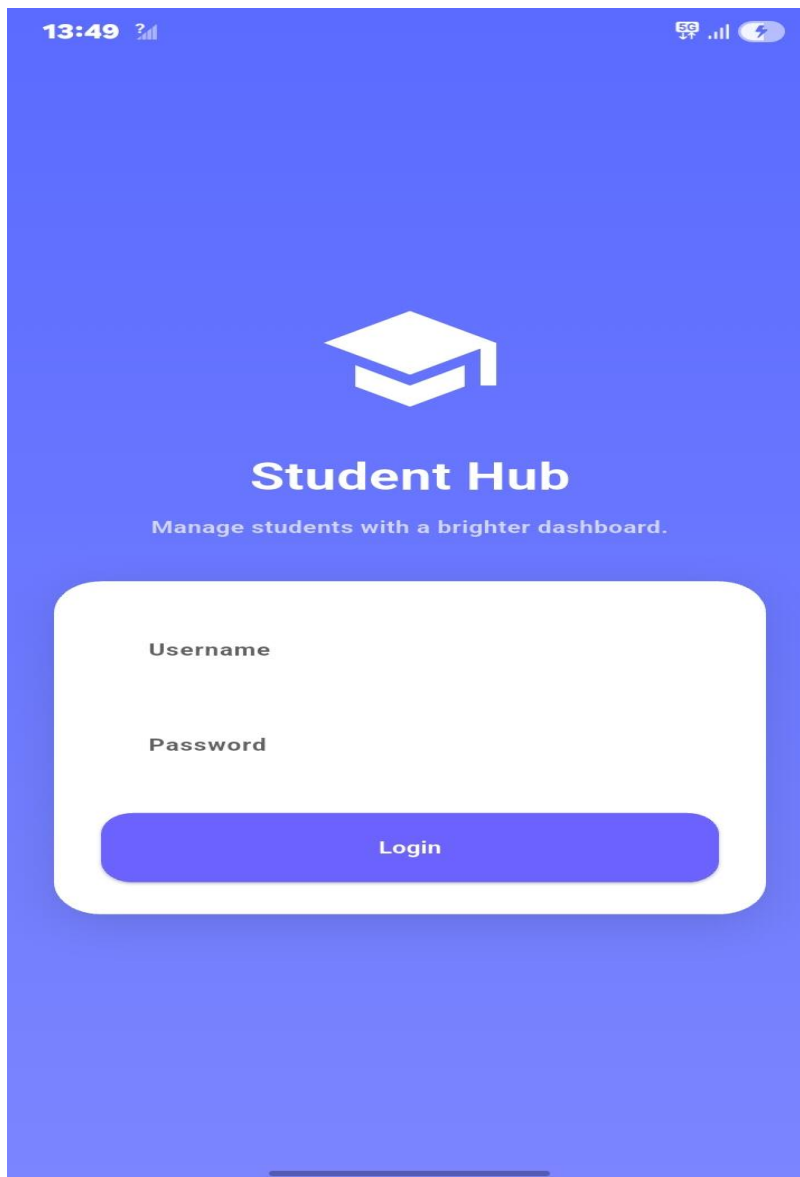
Used Flutter widgets for UI

Forms for data input

Stored data locally

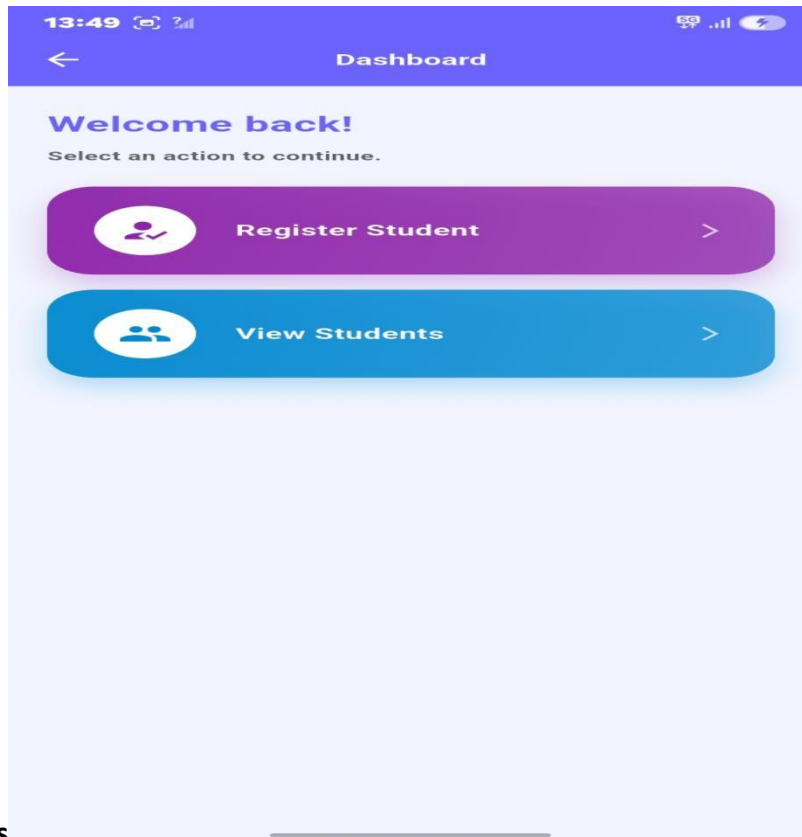
Used Navigator for navigation

7. Running the App



Start emulator or connect phone

Running app on emulator/phone



8. Challenges

Installation errors

Emulator issues

Slow build process

Solution: Used flutter doctor and installed missing components

9. Conclusion

The app was successfully developed and demonstrates basic mobile app development skills using Flutter.